

Question number	Answer	Additional guidance	Mark
1(a)	An explanation that includes two of the following points (in order): - because the surface area needs to be the same (1) - so they have the same {number / amount / substrate} of protein (molecules) (1) - to produce valid {(set of) data / results / experiment / test / comparisons} / for validity (1)	ACCEPT substrate / protein concentration Ignore reliable, accurate, precise, fair test	(2)

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1(b)(i)	water / same solvent (as used to dissolve the enzymes) / buffer	Accept boiled enzyme Denatured enzyme Ignore any qualification of water, e.g., salt water	(1)

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1(b)(ii)	so that any effect can be attributed to the enzymes (and not the solvent) / allows comparisons (1)	ACCEPT as a baseline ACCEPT reverse argument	(1)

Question number	Answer	Additional guidance	Mark
1(c)	(same) temperature and (same) {time / length / duration} (1)	DO NOT ACCEPT pH, size of pieces IGNORE volumes, concentration	(1)

Question number	Answer	Additional guidance	Mark
1(d)(i)	12.5 (cm³)	IGNORE other units than cm ³	(1)

Question number	Answer	Additional guidance	Mark
1(d)(ii)	A description that includes two of the following points: - both enzymes decrease the pH (more than in the control) / at higher concentration the decrease in pH is more than at lower concentration / decrease in pH in bromelain more (than in papain) (1) - bromelain is more effective at breaking down the protein (than papain) (1) - at the higher concentration both enzymes are more effective at breaking down the protein (than at lower concentration) (1)	ACCEPT lower pH {in bromelain than papain / at high concentration than at low concentration} ACCEPT there is a negative correlation between pH and enzyme concentration	(2)

